TAMPINES MERIDIAN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION

CANDIDATE NAME: _____

CIVICS GROUP: _____

H2 MATHEMATICS 9758/02

Paper 2 21 SEPTEMBER 2026

3 hours

Additional materials: Printed Answer Booklet

List of Formulae and Results (MF27)

READ THESE INSTRUCTIONS FIRST

Answer **all** the questions.

Write your answers in the spaces provided in the Printed Answer Booklet. Follow the instructions on the front cover of the Printed Answer Booklet.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you must present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

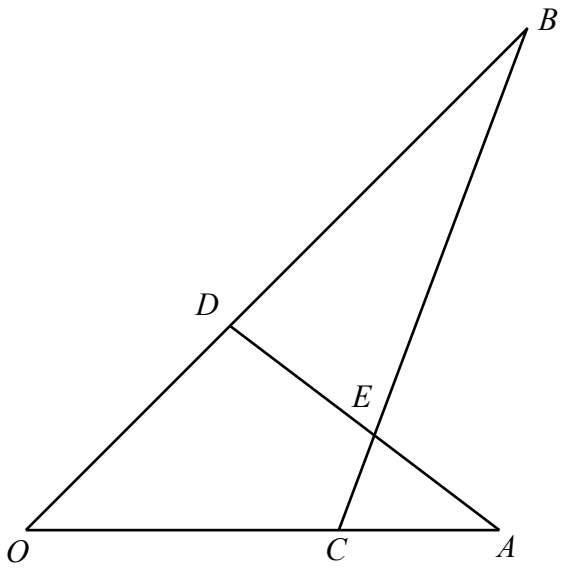
This document consists of 9 printed pages and 1 blank page.

Section A: Pure Mathematics [40 marks]

1 **(a)** On the same diagram, sketch the graphs of $y=|x-a|$ and $y=(x+1)^2$, where a is a real constant such that $a>1$. You should show clearly the axial intercepts of both graphs. [3]

(b) Hence, solve the inequality $(x+1)^2>|x-3|$, giving your answer in exact form. [3]

2



Referred to the origin O, points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. Point C lies on OA such that $OC=kOA$, where $0<k<1$. Point D lies on OB such that $5OD=2OB$ (see diagram above).

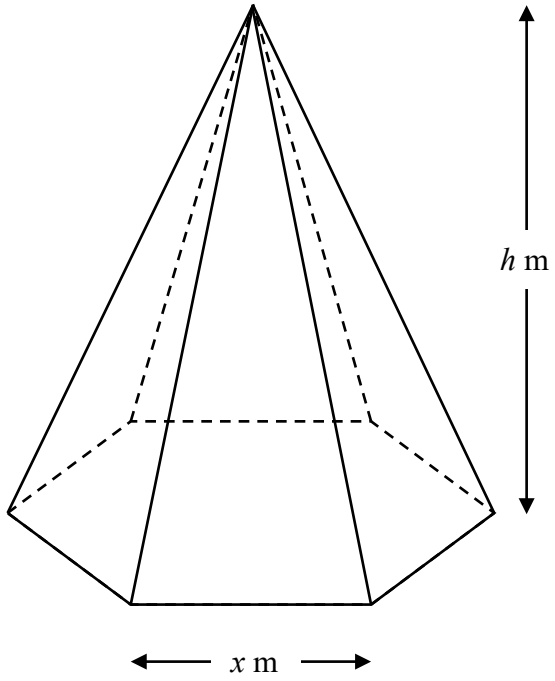
(a) Show that the vector equation of the line AD can be written as $\mathbf{r}=\mathbf{a}+\lambda\left(\frac{2}{5}\mathbf{b}-\mathbf{a}\right)$, where λ is a parameter. [1]

(b) Find, in a similar form, the vector equation of the line BC in terms of k. Hence, find the position vector of point E, where lines BC and AD intersect, expressing your answer in terms of $\mathbf{a}$, $\mathbf{b}$ and k. [5]

(c) Find the area of triangle OEB. Express your answer in the form $\mathrm{f}(k)|\mathbf{a}\times\mathbf{b}|$, where $\mathrm{f}(k)$ is a function of k. [2]

(d) Hence, or otherwise, give the geometrical interpretation of $\lim_{k\to 1}\left[\mathrm{f}(k)|\mathbf{a}\times\mathbf{b}|\right]$. [1]

3



A tentage is designed in the shape of a right pyramid comprising of a regular hexagonal base and six identical triangular surfaces. The pyramid has height h m. Each side of the regular hexagonal base measures x m (see figure above). The interior volume of the tentage, V m^3, is set to be $\frac{75}{2}\sqrt{3}$ m^3. A layer of solar reflective coating is applied to the six exterior triangular surfaces of the rooftop to reflect sunlight and infrared heat away from the shelter. To reduce the amount of coating used, the total surface area of the six exterior triangular surfaces, denoted by S m^2, must be minimised.

(a) Show that $S = \sqrt{\frac{27x^4}{4} + \frac{50625}{x^2}}$. [4]

(b) Using differentiation, calculate the value of x for which S is a minimum and find the corresponding minimum value of S. (You need not prove that the value of S is a minimum.) [4]

[The volume of a pyramid is $\frac{1}{3} \times$ base area $\times$ height.]

4 The function f is defined by

$$\mathrm{f}(x) = \begin{cases} \sin\left(\dfrac{\pi x}{2}\right) + a & \text{for } 0 < x \leq 1, \\ -x + 2 + a & \text{for } 1 < x \leq 2, \end{cases}$$

where a is a positive real number.

It is given that $\mathrm{f}(x) = \mathrm{f}(x+2)$ for all real values of x.

(a) Sketch the graph of $y = \mathrm{f}(x)$ for $0 < x \leq 6$. [3]

(b) Find the exact value of $\int_0^3 \mathrm{f}(x)\,\mathrm{d}x$ in terms of a. [4]

5 The function f is defined by

$$\mathrm{f} : x \mapsto x^2 - 4x + 12 \quad \text{for } x \in \mathbb{R}.$$

(a) Explain, with the aid of a sketch or otherwise, why f does not have an inverse. [2]

The domain of f is now restricted to $x \leq k$, where k is an integer, such that the inverse function f^{-1} exists.

(b) State the greatest possible value of k. [1]

For the remaining parts of this question, take k to be the value found in part **(b)**.

(c) Define f^{-1} in a similar form. [3]

The function g is defined by

$$\mathrm{g} : x \mapsto \ln x, \text{ for } x \in \mathbb{R}, \ 0 < x < 5.$$

(d) Show that the composite function fg exists. [2]

(e) Find the range of fg in exact form. [2]

Section B: Probability and Statistics [60 marks]

6 Four students, two female teachers and one male teacher stand in a line for a photo-shoot.

(a) Find the number of arrangements if there are no restrictions. [1]

(b) Find the number of arrangements in which the two female teachers are together. [1]

(c) Find the number of arrangements in which the three teachers are all separated. [2]

The students and teachers attend a workshop in which they are seated at a round table.

(d) Find the probability that the two female teachers are separated by exactly one student given that the two female teachers are separated. [4]

7 The table below gives the median monthly household employment income per household member, x dollars, and the total life expectancy at birth, y years, in Singapore for certain years from 2003.

x	1337	1448	1736	2129	2503	2799	3033	3621
y	79.1	80.3	81.4	82.1	82.9	83.4	83.2	83.7

Source: Singapore Department of Statistics

(a) Give a sketch of the scatter diagram, as shown on your calculator. State what the scatter diagram suggests about the relationship between x and y. [3]

(b) Find, correct to 5 decimal places, the value of the product moment correlation coefficient between

(i) $\ln x$ and y, [1]

(ii) $\sqrt{x}$ and y. [1]

(c) Using part **(b)**, explain which of the model $y = a\ln x + b$ or $y = c\sqrt{x} + d$ gives a better fit to the data and find the equation of the regression line for this model. [2]

(d) Use the equation of the regression line found in part **(c)** to estimate the total life expectancy at birth when the median monthly household employment income per household is \$5000. Explain whether your estimate is reliable. [2]

8 A food delivery platform TMFoodRide, has two types of delivery riders: motorcycle riders and bicycle riders. Data from peak-hour orders show that

- 65% of peak-hour orders are assigned to motorcycle riders.
- A peak-hour order assigned to a motorcycle rider has a probability of 0.14 to be delivered late.
- A peak-hour order assigned to a bicycle rider has a probability of 0.3 to be delivered late.

(a) Find the probability that a randomly chosen peak-hour order is delivered late. [2]

(b) Given that a peak-hour order is delivered late, find the probability that it was assigned to a bicycle rider. [2]

A regular customer places 14 peak-hour orders over a month on TMFoodRide. Let X be the number of peak-hour orders that are delivered late.

(c) State, in the context of the question, two assumptions needed for X to be well-modelled by a binomial distribution. [2]

It is now given that X can be modelled by a binomial distribution.

(d) Find the probability that at least 3 peak-hour orders are delivered late. [2]

Employees of a company place 10 peak-hour orders on TMFoodRide to be delivered to the company at the same time. Explain why it may not be appropriate to use a binomial distribution for the number of peak-hour orders that are delivered late. [1]

9 A dairy company produces cartons of milk which are labelled as containing a volume of 1000 ml. A consumer group suspects that the volume stated on the cartons is overstated and raises a complaint. A food quality control inspector takes a random sample of 60 cartons and the volume of milk per carton, x ml, are summarised by

$$\sum x = 59760 \qquad \sum x^2 = 59534496 .$$

(a) Find unbiased estimates of the population mean and population variance. [2]

(b) Test, at the 5% significance level, whether the consumer group's complaint is justified. [4]

Following the complaint, the dairy company upgrades its equipment that fills the cartons. The volume of milk per carton now follows a normal distribution with mean 1000 ml and standard deviation 10 ml. The inspector takes a new random sample of 60 cartons. A test at the 5% significance level is carried out to determine if the mean volume of milk per carton is less than 1000 ml.

(c) Find the range of possible values of the sample mean that would lead to the null hypothesis not being rejected. [3]

10 A bag contains 3 blue counters and n red counters where $n \geq 3$. In a game, Mathias removes 3 counters at random from the bag without replacement.

The random variable X, representing Mathias's score, is defined by the number of red counters drawn minus the number of blue counters drawn.

(a) Show that $\mathrm{P}(X=-1)=\dfrac{18n}{(n+3)(n+2)(n+1)}$. [1]

(b) Find, in terms of n, the probability distribution of X. [3]

(c) Find the expectation of Mathias's score, giving your answer in the form $\dfrac{p(n)}{q(n)}$, where $p(n)$ and $q(n)$ are linear. [3]

(d) Given that $\mathrm{E}(X)=2$, find the value of n. [1]

The game organiser later revises the scoring system. Instead of using X, the new score awarded is given by $Y=4X$.

(e) Explain clearly whether you agree with each of the following claims made by the game organiser:

(i) the mean score remains unchanged. [1]

(ii) the standard deviation becomes 4 times larger. [1]

11 A retailer sells both coffee beans and cocoa beans to customers in packets. The masses of packets of beans sold are independent and normally distributed with means and standard deviations as shown in the table.

	Mean (grams)	**Standard Deviation (grams)**
Coffee beans	252	6
Cocoa beans	195	σ

(a) The retailer wishes to take a random sample of the packets of coffee beans from the thousands available on the racks for sale, for quality control purposes. State what it means for a sample to be random in this context. [1]

(b) Given that 10% of the cocoa beans weigh less than 189 grams, find the value of σ. [2]

For the rest of the question, use $\sigma = 4$.

(c) Find the probability that a randomly chosen packet of coffee beans weighs between 240 grams to 250 grams. [1]

(d) The retailer selects packets of coffee beans for distribution. Find the probability that the tenth packet of coffee beans he selects is the third packet that weighs between 240 grams to 250 grams. [2]

(e) Two packets of coffee beans and three packets of cocoa beans are chosen at random. Find the probability that the mean mass of these packets of beans is less than 215 grams. [3]

(f) Coffee beans are sold at \$0.10 per gram and cocoa beans are sold at \$0.05 per gram. Find the probability that the total selling price of 2 randomly chosen packets of coffee beans costs at least \$20 more than 3 times the selling price of a randomly chosen packet of cocoa beans. [4]

(g) A customer bought n packets of cocoa beans. Find the approximate probability that the total mass of cocoa beans bought exceeds $200n$ grams as n becomes very large. [2]

End of Paper